

MLDM Assignment 7

Detailed results for Jandson Ribeiro

Course	Machine Learning and Data Mining WS 2022/23 (copy)
Test	MLDM Assignment 7

This are your test results

Knowledge Questions

 Decision trees

Status

Seen but not answered

Your score

0 / 10

0%

Response

Mark the fowllowing statements about Decision trees to be Right or Wrong.

Note :
Every correct answer gets +2 points and wrong answer gets -1 point. The minimum score of the question is 0 and the negative points are not carried over to the next question.

Unanswered	Right	Wrong	
☐	☐	☐	While building a decision tree, the feature space must be split only into hyperrectangles, whose sides are parallel to the axes.
☐	☐	☐	In Decision trees, classes are sequentially rejected until reaching a final suitable class
☐	☐	☐	After classifying using a Decision tree, the subsets show a higher preference to the corresponding class
☐	☐	☐	At each stage, the decision is made by the question " is feature $x_i = \alpha$? ", where α is any constant
☐	☐	☐	The leaf node belongs to the class , that occurs most often in its parent node

📄

Binary decision tree

Status

Not started

Your score

0 / 7

0%

Response

You can find below the steps of Binary decision tree algorithm. Arrange them in the correct order to obtain the desirable result.

Given ,

X: set of training points, X_t : the root node

x_k : feature , $k : [1,2,...m]$

ΔI : Change in impurity

Label the leaf node with a class

Declare leaf nodes by applying stop splitting rules

Compute change in Impurity $\Delta I(t)$

Choose the feature x and α that produces overall maximum $\Delta I(t)$

For every node in the training set X , different values of α , splitting condition is applied to generate two disjoint subsets

If stop splitting sets do not apply, generate two descendent subsets based on the splitting criteria

Select α leading to the highest $\Delta I(t)$

M

☰

Overfitting

Status

Not started

Your score

0 / 10

0%

Response

What is Overfitting? Explain how it affects Decision Trees. Name any other Classification method that can be used to reduce Overfitting.

0 word

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2/5

True/false

Status

Not started

Your score

0 / 8

0%

Response

Mark whether the statements are right or wrong.

Note :
Every correct answer gets +2 points and wrong answer gets -1 point. The minimum score of the question is 0 and the negative points are not carried over to the next question.

Unanswered	Right	Wrong	
☐	☐	☐	If the Testing dataset is large , the testing error approaches the Generalisation error
☐	☐	☐	If the optimistic error of leaf nodes and parent is the same , and the pessimistic error of parent is higher than its leaf nodes, then the leaf nodes are pruned
☐	☐	☐	Building Random forest involves sampling without replacement.
☐	☐	☐	Random forest is comparatively less sensitive to noise in the training set

 Practice3/5

*** Decision tree pruning

Status

Not started

Your score

0 / 5

0%

Response

A decision tree classifier is used to classify a box of balls to its respective colors. There are 25 balls in total of different colors. Considering there are 3 leaf nodes in the decision tree.

Given , $\lambda = 0.8$.

3 Leaf nodes from the same parent.

Leaf node 1, L1 (Red) : 10 red balls, 2 blue balls

Leaf node 2, L2 (Green) : 5 green balls, 1 red ball

Leaf node 3, L3 (Blue) : 6 blue balls, 1 green ball

In order to prune, calculate its optimistic and pessimistic error.

Give your answer with 2 decimal point values.

Optimistic error :

Pessimistic error :

*** Root node

Status

Not started

Your score

0 / 10

0%

Response

Consider the following dataset. The first four columns are the features and Profitable is the class that we are trying to predict.

price	airbag	capacity	maintenance	profitable
low	no	2	low	yes
low	yes	4	med	no
low	no	4	low	yes
low	no	4	high	no
med	no	4	med	no
med	yes	4	med	yes
med	yes	2	high	no
med	no	5	high	yes
high	yes	4	med	yes
high	yes	2	high	no
high	yes	5	high	yes

Identify which feature is the most suitable feature to be used as a root node while building a Decision tree, using Information Gain measure.

Note that Information Gain values to be filled in the blank are only numbers. Do not add any signs. The answer must have 3 decimal values.

(Information Gain)_{price} =

(Information Gain)_{airbag} =

(Information Gain)_{capacity} =

(Information Gain)_{maintenance} =

Selected root node :

 Programming

 Upload file		
Status	Not started	
Your score	0 / 30	0%
Response Download the file DecisionTrees_ML_7.ipynb and use it in the following task. In the end of the template, a new dataset is provided. Use the generated decision tree d as given in the template (do not change the tree d itself, treat it as read-only) to classify the samples of the new dataset. Decide for each sample in the new dataset, whether rain or no-rain is predicted following the split rules from tree d. For this task, write your code within a user defined function predict_rains() that takes the required input parameters Display the classification result in in your code. Also , include the answer to the following question in the file. Explain the decision tree data structure stored in the variable d. Which python data types are used in which combination? Upload the completed program file. Note : Submission without the upload will not be considered. File No file uploaded		